REVA UNIVERSITY • SCHOOL OF COMPUTING & INFORMATION TECHNOLOGY

AIML-401 • OFFICIAL FACULTY LECTURE MATERIAL

Course Material: RNNs, Vanishing Gradients & LSTM Gating Mechanisms

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1. Primary Teaching Points & Concept Excerpt

“Recurrent Neural Networks model sequential and temporal dependencies through recurrent hidden states. Due to repeated matrix multiplications during Backpropagation Through Time (BPTT), vanilla RNNs suffer severely from vanishing and exploding gradients. Long Short-Term Memory (LSTM) networks resolve this by establishing an uninterrupted cell state highway governed by three multiplicative gates.”

2. Mathematical Formulation & Core Equations

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1. Vanilla RNN Hidden State Update:
    h_t = tanh(W_hh • h_{t-1} + W_xh • x_t + b_h)

2. LSTM Gating Equations:
    Forget Gate:       f_t = σ(W_f • [h_{t-1}, x_t] + b_f)
    Input Gate:        i_t = σ(W_i • [h_{t-1}, x_t] + b_i)
    Candidate State:   C~_t = tanh(W_c • [h_{t-1}, x_t] + b_c)
    Cell State Update: C_t = f_t ■ C_{t-1} + i_t ■ C~_t
    Output Gate:       o_t = σ(W_o • [h_{t-1}, x_t] + b_o)
    Hidden Output:     h_t = o_t ■ tanh(C_t)
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3. 4-Stage Algorithmic Architecture Pipeline

Stage	Pipeline Component	Mathematical / Architectural Operation	Key Objective
1	Sequence Ingestion	Sequential token embedding x_t at timestep t	Input representation
2	Forget Gating	Evaluate f_t ∈ [0, 1] to discard obsolete past memory	Memory purge
3	Information Store	Multiply candidate C~_t by input gate i_t into cell state	Persistent memory update
4	Gated Readout	Scale activated cell state tanh(C_t) by output gate o_t	Emitted hidden state h_t

4. High-Yield Question & Marking Rubric for Midterm IA-2 & SEE Exams

Expected University Exam Question (8 Marks): *'Mathematically explain why standard RNNs fail to preserve long-term context during BPTT. Write all six governing equations for an LSTM cell and explain the exact mechanical role of the forget gate.'*

Official Faculty Marking Rubric:

• 2 Marks: BPTT continuous product expansion and eigenvalue condition for vanishing gradient.

• 4 Marks: Accurate formulation of all 6 LSTM gates and state transition equations.

• 2 Marks: Mechanical role of the cell state additive highway (f_t * C_{t-1}).